

Assignment 1 Report - Sensor Logs and Anomaly Detection

1) Approach

- Generate Node1..Node8 folders and day1..day7 log files in the current path.
- Each log contains 1500 random A-Z status codes.
- Read each file and count code frequencies with Counter.
- Compare each file's count with the average of other nodes times k.

2) Average calculation logic

- For each code and node, sum counts from the other 7 nodes across 7 days.
- $\text{Average} = \text{other_total} / (7*7)$ files.
- If $\text{file_count} > k * \text{average}$, record anomaly.

3) Run results analysis

- Example run with $k=1.3$ and $\text{seed}=42$ produced 22 anomalies.
- Adjust k up/down to keep anomalies between 15-40.

4) Screenshot

- Insert console screenshot of a successful run here.